

NOVEL STATISTICAL METHODS

Performance Metrics for the Comparative Analysis of Clinical Risk Prediction Models Employing Machine Learning

Chenxi Huang, PhD; Shu-Xia Li, PhD; César Caraballo¹, MD; Frederick A. Masoudi², MD, MSPH; John S. Rumsfeld, MD, PhD; John A. Spertus³, MD, MPH; Sharon-Lise T. Normand, PhD; Bobak J. Mortazavi⁴, PhD; Harlan M. Krumholz⁵, MD, SM

BACKGROUND: New methods such as machine learning techniques have been increasingly used to enhance the performance of risk predictions for clinical decision-making. However, commonly reported performance metrics may not be sufficient to capture the advantages of these newly proposed models for their adoption by health care professionals to improve care. Machine learning models often improve risk estimation for certain subpopulations that may be missed by these metrics.

METHODS AND RESULTS: This article addresses the limitations of commonly reported metrics for performance comparison and proposes additional metrics. Our discussions cover metrics related to overall performance, discrimination, calibration, resolution, reclassification, and model implementation. Models for predicting acute kidney injury after percutaneous coronary intervention are used to illustrate the use of these metrics.

CONCLUSIONS: We demonstrate that commonly reported metrics may not have sufficient sensitivity to identify improvement of machine learning models and propose the use of a comprehensive list of performance metrics for reporting and comparing clinical risk prediction models.

Key Words: acute kidney injury ■ machine learning ■ metrics ■ percutaneous coronary intervention ■ precision medicine ■ statistical model

As the interest in precision medicine increases, data scientists are using new methods to produce better predictive models.^{1–3} In this context, the evaluations and head-to-head comparisons of models are growing in importance as these new methods and models emerge.

Machine learning can accommodate nonlinearity and higher order interactions and holds the promise to enhance prediction performance.⁴ This advantage may produce better risk estimates for certain subpopulations,^{5–7} but the commonly reported metrics may miss the subtle improvements. For example, the most reported discrimination metric is the C statistic, also known as AUC-ROC (area under the receiver operating characteristic curve), which summarizes the sensitivity and specificity of a model at all possible risk thresholds. However, models with similar C statistics can have vastly different performance of predicting for a particular risk group.^{8–10}

Other, more recently proposed metrics, to capture changes in individual predictions, such as the net reclassification index, or to quantify clinical usefulness, such as the decision curve analysis, also have limitations.^{11,12} Furthermore, there is a lack of consideration for model implementation, which is an important issue for determining the usefulness of risk prediction models in clinical practice.

Numerous guidelines^{13–16} have been proposed and should be followed for reporting model development and performance evaluation. Recognizing their inadequacy for application to new modeling approaches, efforts are being undertaken to extend such guidelines for studies involving machine learning models, but there is not yet a dedicated study for addressing the issues of performance metrics for these models.^{17,18} Given the need for clinicians to be informed consumers of studies comparing

Correspondence to: Harlan M. Krumholz, MD, SM, Yale School of Medicine, 195 Church St, Fifth Floor, New Haven, CT 06510. Email harlan.krumholz@yale.edu
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model performance, we sought to address the strengths and limitations of established metrics and propose additional metrics, with particular attention to their ability to differentiate performance for various risk groups. Such a suite of metrics provides a comprehensive evaluation and comparison of the models, which is a critical step toward understanding the performance of a model when implemented in practice. Other factors such as the consequences of overtreatment for specific outcomes need to be considered along with the analytical assessment of the models. We provide a working example of comparing 2 risk prediction models for acute kidney injury (AKI) after percutaneous coronary intervention (PCI). We also illustrate the use of a model-agnostic approach to understand discrepancy between models for individual predictions, providing ways to interpret the improvement of the machine learning models.

PERFORMANCE METRICS

Because of the sensitive nature of the data collected for this study, requests to access the data set may be sent to the American College of Cardiology National Cardiovascular Data Registry at cvquality.acc.org.

We focus on metrics that evaluate a model's performance in predicting risks without deriving the effects of risk factors or making causal inference. Our discussion is for models predicting binary outcomes and probabilistic risk estimations; most of the metrics can be applied to other types of models such as survival models or equivalent metrics that exist, but extra consideration is needed. The metrics included (Table 1) are also model agnostic, providing a general guideline for models developed by various techniques. These metrics aim to quantify the fidelity of the probabilistic predictions to the true risks of the patients.

Metrics of model implementation are also included because the models developed by more advanced techniques, while likely having improved performance, may incur more resources or difficulties in implementation. Besides the metrics, proper assessment and comparison of model performance also rely on the training and validation strategy, considerations of the data upon which the model is built and metrics are calculated.^{19–21} For this article, however, we focus on performance metrics. The analysis was approved by the Institutional Review Board at the Yale University.

Overall Performance

The most commonly reported overall performance metric is the Brier score (also known as the loss or mean-squared-error of the predictions), defined as the mean squared difference between the predictions and the observed outcome.²² This score is a proper scoring rule for predicting binary outcomes, meaning that a better

model always dominates in Brier score.²³ When evaluated on the same data, a smaller Brier score indicates better performance. The Brier scores calculated on 2 different data sets should not be compared directly, since the value of the Brier score depends on both the prevalence of the event in the data and the performance of the model. In rare-event cases, the Brier score can be misleadingly low for even noninformative models such as a model producing 0 risk for everyone.

The Brier score does not have a set value or range for denoting it as good or bad. To overcome the limitations of interpreting the numerical values of the Brier score, we propose the use of the Brier skill score²⁴ in conjunction with the Brier score. The Brier skill score reflects the skill gained in the model for prediction compared with a model with no skill (ie, a model that cannot differentiate risks). A model that is useful for clinical decision-making should have a Brier skill score of at least above zero. The Brier skill score using a no-skill model that predicts the event prevalence for all patients is equivalent to the scaled Brier score.²⁵ A Brier skill score normally ranges from 0 to 1 and is shown as a percentage, with a score of 0% showing no skill and a score of 100% showing perfect prediction. The Brier skill score is more intuitive and more sensitive to performance differences than the Brier score.^{26–28}

The Brier score or the Brier skill score should not be used alone for comparing model performance. They are composed of several domains of performance, including discrimination and calibration, which should also be reported.^{29,30} Models with similar Brier scores may result in meaningfully different discrimination and calibration performance. Borrowing notions from climate research, we introduce components of the Brier score, including a resolution term that is related to discrimination, 3 reliability terms for calibration, and a term that is solely determined by the outcome prevalence (Methods in the [Data Supplement](#)).^{31,32} This decomposition is dependent on a binning strategy that groups patients by predictions. The most common binning strategy uses the deciles of the predictions, which is used in calculating the Hosmer-Lemeshow goodness-of-fit test for calibration.³³ For the included example, we illustrated the use of 2 binning strategies, the deciles of predictions, and equidistant 10 bins between 0 and 1.

In addition, we propose the use of the prediction squared error, which differs from the Brier score in that it accounts for the random component of the outcome for predicting binary events rather than predicting probabilistic risks. Usually, binary event predictions are derived by thresholding the estimated risks. However, this approach ignores the stochastic (eg, the irreducible error) variation for individuals and underestimates the prediction error.³⁴ We calculate the predicted binary events by random sampling from a Bernoulli distribution with probability equal to the estimated risks. The prediction squared error metric is calculated as the mean squared difference

Table 1. List of Performance Measures for Evaluating Comparative Effectiveness of Risk Prediction Models

Property	Summary statistic	Indicator of better performance	Visualization, if any	Explanation
Overall performance	Brier score	Lower		Error of probabilistic predictions from the binary outcomes. There is no set value for a good Brier score; values should be compared between models or used in conjunction with the Brier skill score.
	Brier skill score	Higher		Relative accuracy of probabilistic predictions compared with a no-skill model. A model useful for clinical decision-making should have a Brier skill score above 0.
	Prediction squared error	Lower		Error of binary predictions from the binary outcomes. This metric captures the uncertainty for decision-making based on predictions of whether patients will experience the outcome.
Discrimination	AUC-ROC	Higher	ROC curve	The TPR and FPR at various thresholds on predictions. The value may not reflect key differences at important thresholds for decision-making; the curve should be visualized to compare models.
	AUC-PRC	Higher	PRC curve	The precision and recall at various thresholds on predictions. The value may not reflect key differences at important thresholds for decision-making; the curve should be visualized to compare models.
Reliability/calibration	Calibration slope and intercept	Slope closer to 1 and intercept closer to 0	Reliability/calibration plot	Overall agreement between predictions and observed outcomes. This metric may miss important miscalibrations in subgroups and need to be used in conjunction with the components of reliability.
	Reliability-deviation	Lower		Error of predictions from the observed risks across risk bins. This metric reflects the calibrations across the risk groups and is important for interventions targeting different risk groups, where under- or overtreatment may result from under- or overestimation of risks in these risk groups.
	Reliability-within-bin variation	Lower		Variation of predictions within each risk bin. This metric reflects the spread of predictions for patients within risk groups. A larger spread means that there are patients whose risks may be much lower or higher than the average and intervention targeting this risk group may not be suitable for these patients.
	Reliability-within-bin covariance	Higher		Accuracy of predictions' ranking according to outcomes within each risk bin. This metric reflects the accuracy of ranking patients' risks within risk groups. A large value may indicate the potential of finer scale risk stratification for more personalized treatment.
Resolution	Resolution	Higher	Reliability/calibration plot	Separation of risk bins defined by predictions. This metric reflects the range of risk groups the model can identify. A larger range indicates the ability of identifying more extremes of risk groups for prioritizing interventions.
	Predictive range	Higher		Separation between the lowest and the highest risk groups defined by predictions
Reclassification	RMSE via neighborhood estimate	Lower	Reclassification scatter plot	Improvement in risk stratification accuracy from predictions by a new model compared with an existing one. There is no set value for a good reclassification metric; values should be compared between models and along with the scatter plot. Interventions targeting patients with specific risk ranges need to consider from the plot whether these patients had improved risk estimates.
Implementation considerations	No. of variables			How difficult the extraction of variables is.
	Computation resources required			How difficult the calculation of the risk is.

AUC-PRC indicates area under precision-recall curve; AUC-ROC, area under the receiver operating characteristics curve; FPR, false positive rate; PRC, precision-recall curve; RMSE, root-mean-squared error; ROC, receiver operating characteristics curve; and TPR, true positive rate.

between binary predicted and observed outcomes, and lower prediction squared errors indicate better models. This metric captures the uncertainty for clinicians making treatment decisions for individual patients based on whether they will experience the outcome. The mean and standard deviation of the prediction squared error can be calculated by multiple trials of random sampling.

Another commonly used overall performance metric for binary outcomes is the Logarithmic score (often known as the log-loss). Similar to the Brier score, the log-loss measures how close the predicted probabilities are to the observed binary outcome but penalizes errors differently. Compared with the Brier score, the log-loss increases more rapidly for extreme values of predictions around 0 or 1 that are untrue. The choice between the

Brier score and log-loss is dependent on the importance of these cases for certain clinical applications. However, given that log-loss is often implemented as the loss function for deriving the models,³⁵ such extreme errors are usually rare. In this study, we focus on the discussion of the Brier score because it is generally similar to the log-loss without the extreme errors and more importantly, it can be further decomposed into calibration and discrimination components that help better understand the difference between models.

Discrimination

The ability of the model to distinguish between those with and without the outcome is called discrimination. The most

commonly reported discrimination metric is the AUC-ROC.³⁶ The receiver operating characteristic curve is a 2-dimensional plot of the true positive rates (the proportion of the accurately predicted events in all observed events) against the false positive rates (FPRs; the proportion of the predicted events in all observed nonevents).³⁷ The true positive rate and FPR of each point on the receiver operating characteristic curve plot are calculated by thresholding the predictions to produce binary predictions. Of note is that the true positive rate and FPR themselves have limited usefulness since their values are threshold specific and the threshold depends on particular applications.

The AUC-ROC should not be compared for models evaluated on different datasets. For low-prevalence cases, the AUC-ROC can be insensitive to changes in discrimination because the FPR is overwhelmed by the large number of nonevents and thus does not drop drastically as the number of false positives increases.^{38,39} An alternative metric is the precision and recall curve (PRC). Precision is also known as positive predictive value and is the proportion of accurately predicted events (or true positives) in all positive predictions; recall is also known as sensitivity or true positive rate. Precision is more sensitive than FPR as it is not affected by the number of nonevents. Summary statistics of the area under the PRC (AUC-PRC) can be calculated.⁴⁰ However, these summary statistics do not differentiate curves with different shapes.⁹ This means that 2 models with similar AUC-ROCs can have vastly different sensitivity and specificity for identifying certain risk groups. The receiver operating characteristic curve and PRC of the models should be compared visually alongside the summary statistics. In addition, curves that plot other pairs of accuracy metrics such as negative predictive value versus specificity may also be useful in evaluating the clinical utility of specific models.

There are other popular overall performance metrics that are not included in this study, such as the F score and the net benefit.⁴¹ Both of these metrics use weighting strategies to summarize 2 metrics to reflect the tradeoff between them. F score weighs between precision and recall while the net benefit weighs between true positives and false positives. The net benefit is especially helpful for determining the usefulness of a model to administer treatment based on the predicted risks. However, the desired weights may depend on the particular outcome and may even differ among multiple risk groups.⁴² In such cases, clinical input is required when calculating the F score and the net benefit, and different conclusions of model comparison may result from the use of different weights.

Reliability/Calibration

Reliability, more commonly known as calibration, evaluates the agreement between the probabilistic predictions and true risks. Poor calibration can lead to either

under- or overestimation of risks, which in turn may result in under- or overtreatment.

As one of the components of the Brier score, reliability further includes 3 subcomponents (Methods in the [Data Supplement](#)). The first subcomponent, referred to as deviation, calculates the average squared difference between the event rate and average prediction of patients in each risk bin. This metric reflects the calibrations across the risk groups, that is, whether the average prediction for patients in the risk group is equal to the observed event rate of these patients. Such calibration is important when different interventions are targeting different risk groups; under- or overestimation of risk for any risk group may result in under- or overtreatment. The second subcomponent, referred to as within-bin variance, calculates the average variance of predictions for patients within each risk bin. This metric reflects the spread of predictions for patients within risk groups. A larger spread indicates there are patients whose risks may be much lower or higher than the average risk of that risk group. What this means is that these patients may not be suitable for the treatment that is targeting this risk group. Therefore, this metric is important to consider with the consequences of under- or overtreatment for a specific intervention. The last subcomponent of reliability, referred to as within-bin covariance, calculates the average covariance of predictions and observed outcomes within each risk bin. This metric reflects the accuracy in risk ranking the patients by the predictions within each risk group. A large value of this metric may indicate that the risk groups can be further divided without losing too much in accuracy of risk stratification. This is desired if potential finer scale of risk groups might be beneficial for more personalized treatment.

Reliability can be visualized by calibration plots with the average predictions in risk bins on the x axis and the event rate within these bins on the y axis (Figure 1 in the [Data Supplement](#)). Deviation can be visualized by the vertical distances of the event rates in risk bins from the 45° diagonal line. Another common calibration metric is the regression slope and intercept of the observed outcomes versus predictions, the calculation of which does not require binning, and a slope closer to 1 and an intercept closer to 0 indicates better calibration.⁴³ One can also derive nonparametric fit between the outcomes and the predictions, capturing potential nonlinearity in the relationship.⁴⁴ The calibration slope has several limitations. First, models with calibration slopes close to 1 may under- or overestimate risks in subgroups. Second, calibration slopes fail to account for distribution of risk predictions for patients within different risk groups. Therefore, we propose to use all of the subcomponents for reliability in addition to calibration slope. As models are increasingly used for precision medicine, model performance for subgroups is particularly important in addition to performance for the overall population.

Resolution

As the other component of the Brier score, resolution is calculated as the average squared difference between the event rate within each risk bin and the population event prevalence. Resolution is closely related to discrimination; resolution measures whether the expected outcome differs as the prediction changes, whereas discrimination measures whether the prediction differs given different outcomes. The resolution metric has a more intuitive clinical interpretation than AUC-ROC: it reflects the range of risk groups that the model can identify. A larger range may indicate that the model could help target more extremes of risk groups for prioritizing interventions and better allocation of resources. An alternative resolution metric is the predictive range, calculated by the difference between the event rates in the first and last risk bins.

Resolution and reliability are equally important. A model with high resolution but low reliability cannot be trusted for risk prediction, while a model with good reliability but low resolution is not useful for identifying patients from different risk groups. Further, since resolution and reliability are components of the Brier score that depends on the model, they present a clear picture of what aspects of performance contribute to difference in overall performance. Resolution can also be visualized in the calibration plot by the vertical distances between the event rates in risk bins from the population prevalence or between the first and last risk bins (Figure I in the [Data Supplement](#)).

Reclassification

Reclassification assesses how patients are reclassified by a second model from the first model and the appropriateness of the reclassification. The evaluation of reclassification is helpful for interpreting the usefulness of a new model, especially when only a small subgroup of patients is reclassified, which usually results in minimal improvement in the C statistic.^{45,46} While being a small number, the changes in risk predictions for these patients could be important if they are part of the target group to receive risk mitigation strategies. The net reclassification index and the integrated discrimination improvement are 2 common reclassification metrics that calculate the proportion of patients having improved predictions by the new model.⁴⁷ However, net reclassification index may be misleading since a new model with additional uninformative predictors can still produce positive net reclassification index.^{11,48} Moreover, these summary statistics cannot assess the difference of the models in predicting risks for an individual patient.

We propose to visualize the reclassification of individual patients' risks via a 2-dimensional scatter plot. Each point on this plot represents a patient with the predictions

from the two models on the two axes. To evaluate the appropriateness of the reclassification, the scatterplot is colored by the observed risks, estimated by calculating the event rate among those in the neighborhood of that patient on the scatter plot. This plot reflects whether and how much improvement in risk prediction for each patient is produced by the new model over the existing model. Interventions that target patients with specific risks need to consider from the plot whether these patients achieve improved risk predictions. We can also calculate the root-mean-squared error between the predictions and the estimated observed risks as the summary statistic.

Implementation

It is critical to consider the potential barriers in implementation when comparing prediction models. Some of the issues are specific to the site of implementation, such as the infrastructure of the health system. These barriers are important to consider but are not related to the models. Another potential barrier is the missingness in the data; several strategies are available and applicable to all models.

We propose to include at least 2 metrics that are closely related to the prediction models. The first is the number of variables included in the model. While more variables may improve performance, these variables may not be equally accessible and their extraction from clinical notes or laboratory results may increase costs and time. The second metric is the computation resources needed. Logistic regression models can be converted to integer-based risk scores, producing risks without a calculator.⁴⁹ With the increasing number of models that use tree-based methods and neural networks, electronic devices are needed to produce risk predictions. While developing and maintaining these advanced models could be computationally expensive, the bedside calculation for producing real-time predictions is usually instantaneous and not an obstacle for implementation.

Interpreting Model Difference

Understanding why the new model reclassifies patients from the existing model helps to identify where the improvement results from and whether it makes clinical sense. This is especially important for a black-box machine learning model for which interpretation is not as simple as in regression models. There are various methods for interpreting machine learning models; some require knowledge of and access to the underlying structure of the model or additional modeling and computation. Considering the need for clinical applications to have a timely and easy-to-implement mechanism for interpretation, we examined the utility of a model-agnostic approach to understand the differences in predictions for the same patient. This approach is

relevant to the counterfactual methods.⁵⁰ For a patient with different predictions by 2 models, we change the value of each variable to other alternative values and observe the changes in both predictions to see whether the discrepancy is resolved. Through this approach, key variables driving differences in the risk predictions can be obtained to provide information on the underlying reasons for reclassification.

COMPARATIVE ANALYSIS OF MODELS PREDICTING AKI

We used 2 models predicting the risk of AKI after PCI to demonstrate the use of proposed metrics for comparative analysis of their performance. One (model 1) uses 13 variables (Table 1 in the [Data Supplement](#)) and logistic regression with no interaction terms, and the other (model 2) is a published model using the same 13 variables and employing gradient descent boosting.⁶ The details of the data and model development are described elsewhere,⁶ but briefly, the models were developed using data from the National Cardiovascular Data Registry CathPCI Registry including PCIs between June 1, 2009, and June 30, 2011. The comparison of model 1 and model 2 was performed on PCIs between July 1, 2011, and June 30, 2017. The analysis was approved by the Institutional Review Board at the Yale University.

Patients

A total of 947 091 PCI patients were included in the training set, with an average age of 64.8 years, 32.8% women, and an AKI rate of 7.4% (Table II in the [Data Supplement](#)). The validation set consisted of 306 385 patients with an AKI rate of 7.5%. Compared with patients in the training set, those in the validation set were older (65.3 years) and less often female (31.9%).

Performance Comparison

The results of comparing model 1 and model 2 in predicting post-PCI AKI risks are summarized in Table 2.

Overall Performance

Model 1 produced a higher Brier score than model 2 (0.0632 versus 0.0621; Table 2), showing a higher prediction error. Further, model 2 had a higher Brier skill score than model 1 (11.3% versus 8.4%), showing larger improvement over the no-skill model. In addition, from random draws of 30 independent trials, model 1 had a mean (standard deviation) prediction squared error of 0.1338 (0.0002), which is higher than model 2 (0.1297 [0.0001]).

Discrimination

Model 2 showed slightly higher AUC-ROC than model 1 (0.773 versus 0.749; 3.2% increase), but the percentage improvement is more prominent in AUC-PRC (0.281

Table 2. Performance Comparison by Summary Statistics Between Model 1 (Logistic Regression) and Model 2 (XGBoost)

Property	Binning strategy	Metric	Model 1	Model 2	%Improvement*
Overall accuracy	NA	Brier score	0.0631	0.0612	3.1
		Brier skill score	8.4%	11.3%	34.1
		Prediction squared error, mean (standard deviation)	0.1338 (0.0002)	0.1297 (0.0001)	3.1
Discrimination	NA	AUC-ROC	0.749	0.773	3.2
		AUC-PRC	0.238	0.281	18.1
Reliability/calibration	NA	Calibration slope	1.074	1.071	4.4
		Calibration intercept	0.013	0.002	85.5
	Deciles	Deviation ($\times 10^{-4}$)	1.1	0.9	15.8
		Within-bin variation ($\times 10^{-4}$)	13.0	15.9	−22.6
		Within-bin covariance ($\times 10^{-4}$)	21.0	30.0	43.3
	Equidistant bins	Deviation ($\times 10^{-4}$)	1.5	1.0	35.3
		Within-bin variation ($\times 10^{-4}$)	5.3	5.3	1.1
		Within-bin covariance ($\times 10^{-4}$)	9.7	9.9	2.2
Resolution	Deciles	Resolution	0.0051	0.0064	26.4
		Predictive range	24.6%	27.9%	13.4
	Equidistant bins	Resolution	0.0055	0.0074	34.2
		Predictive range	47.7%	85.5%	79.2
Reclassification	250 nearest neighbors	Root-mean-squared error	0.0483	0.0195	59.7
	5% difference neighbors	Root-mean-squared error	0.0470	0.0160	66.0
	10% difference neighbors	Root-mean-squared error	0.0469	0.0160	65.8

SD AUC-PRC indicates area under precision-recall curve; AUC-ROC, area under the receiver operating characteristics curve; and NA, not applicable.
*%Improvement is referring to improvement of model 2 compared with model 1. Negative values of %improvement indicate worse performance of model 2 compared with model 1.

versus 0.238; 18.1% increase), indicating that in this use case, the AUC-PRC shows higher sensitivity to changes in discrimination than the AUC-ROC. The receiver operating characteristic curve and PRC plots further revealed that the discrimination of model 2 is better than model 1 at any threshold (Figure 1A). PRC plot showed that model 1 had incorrectly predicting patients with non-events to be high risks (precision decreased from 1 to 0.5 and then increased to 0.7 within a small range of high thresholds with recall around 0.).

Reliability/Calibration

Using deciles of predicted risks, model 2 had better reliability than model 1 except for within-bin variation. Model 2 had a 15.4% lower deviation and 43.3% higher within-bin covariance than model 1, both demonstrating that model 2 had better calibration in the risk groups. Model 2, however, had a 22.6% higher within-bin variation than model 1. This larger spread of patients in the risk bins indicates that model 2 is less accurate when approximating patients' risks in the risk bins by the average prediction. Using the equidistant bins instead, model 2 showed better reliability than model 1 in all 3 subcomponents.

As shown in Figure 1B, the two models had similar calibration plots when evaluated in deciles of predicted risks. Model 2 had a calibration intercept closer to 0 compared with model 1 (0.002 versus 0.013; 85.5% improvement) and a calibration slope slightly closer to 1 (1.071 versus 1.074; 4.4% improvement). The deviation

and within-bin covariance showed a larger percentage improvement than the calibration slope. This conveys that while the models appear similar in overall calibration performance, they may be different in performance for individual patients in risk groups. When evaluated in equidistant bins, model 1 overestimated risks in higher risk groups, indicated by larger deviation from the diagonal line for patients with closer to 1.0 predicted risks.

Resolution

In both binning strategies, model 2 showed higher resolution than model 1 (Table 2). This was also revealed in the calibration plot, where the first and 10th risk bins were further apart in observed event rates for model 2 compared with model 1 (Figure 1B).

Reclassification

The reclassification scatter plot (Figure 2A) compared predictions by 2 models and was colored by the observed rate in a 5%-radius neighborhood. While the majority of the patients (Figure 2B) were around the diagonal line of the scatter plot indicating similarity between the two models, there were a few hundred high-risk patients (indicated by the red color) in the upper-left corner of the plot who were correctly estimated by model 2 as high risk but underestimated by model 1. Further, a small number of low-risk patients close to the lower right corner were overestimated by model 1.

Consistent with the scatter plot, the root-mean-squared error of reclassification error was lower for

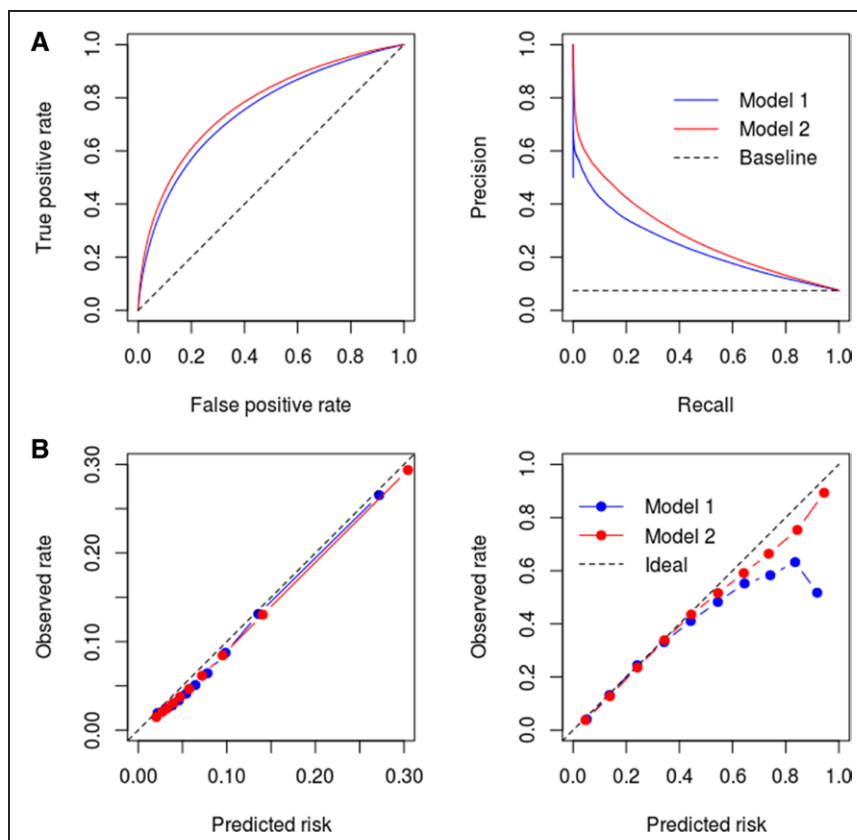


Figure 1. Visualization of model discrimination and calibration. Model comparison in (A) receiver operating characteristics curve (left) and precision-recall curve (right) curves and (B) calibration in deciles (left) and equidistant bins (right).

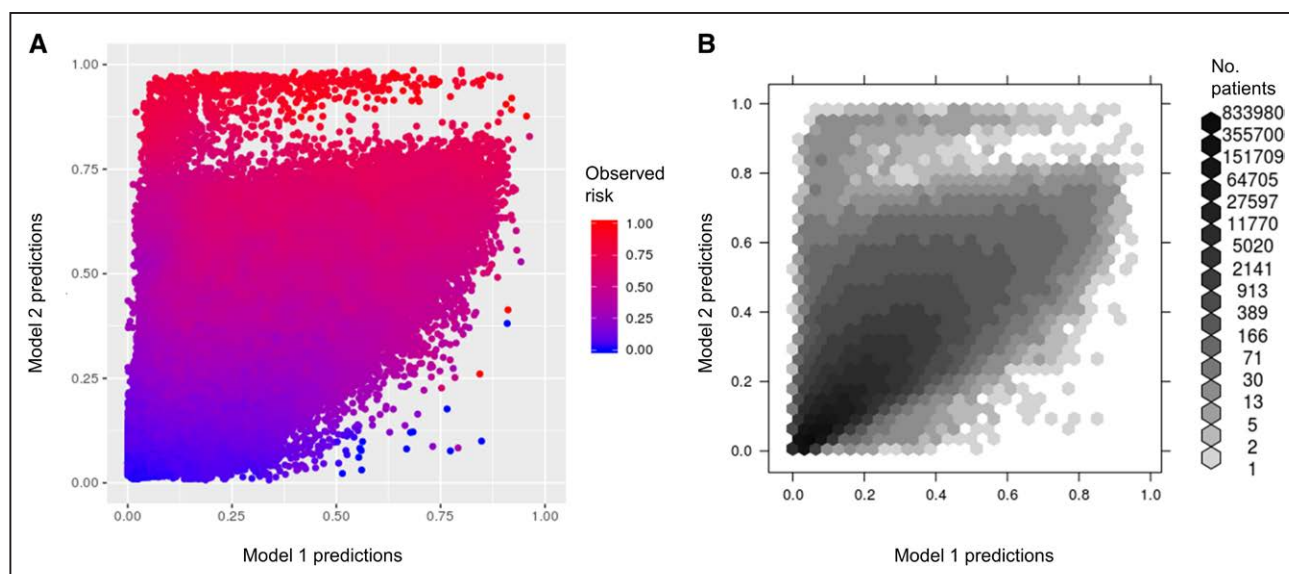


Figure 2. Reclassification scatter plot of model 1 predictions (x axis) versus model 2 predictions (y axis), colored by observed rate in neighborhood of within 5% difference in Euclidean distance in the two predictions (A) and density of the reclassification plot (B).

model 2 compared with model 1 (Table 2), showing that the reclassification of patients by model 2 moved patients to more accurate risk estimates than they were predicted by model 1.

Implementation

Models 1 and 2 use the same variables, so there is no difference in the accessibility of the variables between the two models. In terms of computational resource, model 1 can be converted into an integer-based risk score to be used without a calculator while model 2 requires an electronic device to store the model and calculate the risk.

Interpreting Risk Reclassification

We examined 2 patients with different predictions by the two models. Patient 1 had AKI and a 5.4% predicted risk by model 1 compared with a 70.7% risk predicted by model 2. Figure IIA in the [Data Supplement](#) shows that changing the value of the estimated glomerular filtration rate for this patient resolved the discrepancy between the models. Looking closely at the profile of this patient (Table III in the [Data Supplement](#)) revealed an estimated glomerular filtration rate higher than the last quantile of the cohort. This patient was considered low risk by model 1 due to the high estimated glomerular filtration rate, but this patient still had an increase in creatinine that qualifies for a diagnosis of AKI. For patient 2, who did not have AKI, model 2 had a lower predicted risk than model 1 (10.4% versus 48.3%). Figure IIB in the [Data Supplement](#) revealed that preprocedural hemoglobin is the key variable for this discrepancy. This patient had a preprocedural hemoglobin lower than the first quantile of the cohort and was thus considered high risk by model 1 (Table IV in the [Data Supplement](#)). These two examples implied that model 2 is able to detect nonlinear

relationships between estimated glomerular filtration rate or hemoglobin and AKI for some subgroups.

SUMMARY

We have provided detailed discussion on the limitations of commonly reported metrics for model performance comparison involving machine learning models. The C statistic and calibration slope—the most published metrics for discrimination and calibration—do not have sufficient sensitivity for capturing the advantage of machine learning models, especially when the improvement is for subsets of the population. Additional metrics, including the components of the Brier score and visualization of the reclassification accuracy, need to be considered when comparing models.

The implementation of the model can also face the challenge of missing values. When developing the risk prediction model, the variables that are missing in the training data are often imputed.⁵¹ The same imputation strategy could potentially be used at model implementation. Alternatively, an additional variable indicating missingness can be included as one of the predictors.^{52,53} The amount of missingness in clinical data might be higher for racial or ethnic minorities, which may result in the model performing worse for these subgroups.⁵⁴ In addition to imputation strategies, efforts should also be made to examine in such cases whether the models perform significantly worse in minority groups, using the proposed metrics.

There are other factors to consider when comparing models, in addition to the analytical assessment of performance. Whether the numerical difference presented in such assessment through the various metrics is clinically

meaningful depends on the specific outcome, targeting population, and costs of implementation. When a model shows better performance in some metrics but is lacking in others, one would need to consider what metrics are important for the particular outcome and weight the benefit and risk of choosing one model over another.

Further, it is worthwhile to consider prespecifying metrics that are most important to the clinical application under study. This approach is consistent with the standard protocol to prespecify the primary outcome for clinical trials. Such practice provides guidance on interpreting the results of evaluation and comparative analysis of prediction models. It may also be possible to develop models that directly optimize certain metrics. Nevertheless, we recommend reporting these different metrics to identify any potential deficiency in performance not measured by the prespecified metrics.

Validation strategy is equally important as the metrics for producing a reliable conclusion about model performance comparative analysis. The goal of a validation strategy is to accurately estimate the performance of the model on unseen data for prospective use.²¹ One should not evaluate the performance on the data used to develop the model, unless using optimism correction methods, to prevent overestimating the model performance.⁵⁵ Validating the model on unseen data is especially important for machine learning models, which tend to have a large number of parameters that can easily overfit the training data.^{56,57} The two most commonly used validation strategies are cross-validation and bootstrapping. These strategies rely on specific data splitting or resampling methods to create test sets that mimic unseen data.⁵⁸ There has also been an emphasis in reporting studies of prediction models to include results on external validation data, in addition to internal validations by cross-validation or bootstrapping.^{59,60} Internal validation examines the performance of the model for a population with similar characteristics to those in the training data, while external validation examines whether the model could be extended for use in a slightly different population. However, fully independent external validation data may not always be available. Alternative strategies that divide data in a nonrandom way, such as by site or time, have been proposed when external validation data are lacking.⁶¹ Furthermore, models applied to new data sets may need updating when new data sets have different characteristics or outcome prevalence than the training data. Several recalibration approaches have been proposed to update models under these scenarios.^{6,62}

Our study has the limitation that the metrics we proposed for reporting comparative analysis of clinical prediction models were not derived through a consensus process but rather are based on literature review and experience of the authors. Future investigation is needed to assess the perception of these metrics from

researchers in developing risk prediction models and the impact of their use in improving the adoption of models in clinical practice.

We have clarified the issues of commonly reported discrimination and calibration metrics for applications to risk prediction models using machine learning techniques and proposed the use of additional metrics to capture improvement from the machine learning models. These metrics can be applied to models developed by logistic regression and advanced machine learning techniques. Together, these metrics provide a full picture of model performance.

ARTICLE INFORMATION

Affiliations

Center for Outcomes Research and Evaluation, Yale New Haven Hospital, CT (C.H., S.-X.L., C.C., H.M.K.). Division of Cardiology, University of Colorado Anschutz Medical Campus, Aurora, CO (F.A.M., J.S.R.). Ascension Health, St Louis, MO (F.A.M.) Department of Internal Medicine, University of Missouri, Kansas City, MO (J.A.S.). Department of Cardiovascular Medicine, Saint Luke's Mid America Heart Institute, Kansas City, MO (J.A.S.) Department of Biostatistics, T.H. Chan School of Public Health, Harvard University, Boston, MA (S.-L.T.N.). Department of Health Care Policy, Harvard Medical School, Boston, MA (S.-L.T.N.). Department of Computer Science & Engineering, Texas A&M University, College Station (B.J.M.). Department of Health Policy and Management, Yale School of Public Health New Haven, CT (H.M.K.). Section of Cardiovascular Medicine, Department of Internal Medicine, Yale School of Medicine, New Haven, CT (H.M.K.).

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Supplemental Materials

Supplementary Methods
Figures I and II
Tables I–IV

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